Modeling Network Growth - The Barabási-Albert (BA) Model

Part 1: Implementation of the Barabási-Albert Model

1.1. Algorithm Review

The BA model generates a network through the following steps:

1. **Initialization:** Start with a small connected network of m_0 nodes (e.g., a complete graph).
2. **Growth:** At each time step, add a new node to the network.
3. **Preferential Attachment:** Connect the new node to m, $(m \leq m_0)$ existing nodes. The probability that the new node connects to an existing node i is proportional to its degree k_i:

$$P(k_i) = \frac{k_i}{\sum_j k_j}$$

1.2. Implementation from Scratch

Write a Python function `ba_model_scratch(N, m, m0)` that generates a BA network.

1. Initialize a graph as a list of lists or using a NetworkX graph. Start with a complete graph of m_0 nodes.
2. For each new node from m_0 to $N - 1$:
   a. Calculate the attachment probability for every existing node.
   b. Select `m` distinct nodes to connect to, using a weighted random selection based on the probabilities. *(Hint: Use* `np.random.choice` *with the* `p` *parameter).*
   c. Add the new node and create the `m` links.
3. Return the generated graph.

Challenge (Optional): Can you implement the "linearized" version of the algorithm that is more efficient for large N?

1.3. Implementation using NetworkX

Use the built-in function to generate a BA graph for comparison.

Part 2: Empirical Analysis of the BA Model

Generate a single BA network with $N = 10000$ nodes and $m = 2$ using your `ba_model_scratch` function (or the built in function in `NetworkX` for faster computation if your scratch implementation is slow.)

2.1. Visualizing the Network

Create a visualization of the first **100** nodes added to your network. Use a spring layout and set the node size proportional to its degree. Observe the presence of high-degree hubs.

2.2. Analyzing the Degree Distribution

The key signature of a scale-free network is a power-law degree distribution:

$$P(k) \sim k^{-\gamma}$$

where γ is the degree exponent. For the BA model, the theoretical value is $\gamma = 3$.

1. Calculate the degree for each node.
2. Plot the **Probability Mass Function (PMF)**, $P(k)$, on a log-log scale.
3. Plot the **Complementary Cumulative Distribution Function (CCDF)**, $\mathrm{P(K>k)}$, on a log-log scale. The CCDF is often cleaner and less noisy for tail analysis.
4. Visually inspect the plots. Do the central parts of the distributions appear linear on the log-log scale?

2.3. Estimating the Power-Law Exponent (γ)

Method 1: Linear Regression on the CCDF

1. On the log-log CCDF plot, perform a linear regression on the linear portion of the curve to fit a line.
2. The slope of this line, α, is related to $\gamma = 1 - \alpha$. Report your estimated γ.

Method 2: Using the `powerlaw` Python Library

1. Install the `powerlaw` package.
2. Use it to fit a power law to your degree data and report the estimated γ. Compare this result with your linear regression estimate and the theoretical value of 3.

Part 3: Validation with a Real-World Network

3.1. Dataset and Preprocessing

Download a real-world network dataset. A good example is **Citation Network (ArXiv GR-QC)** dataset from SNAP (SNAP: CA-GrQc).

This is a collaboration network of co-authorships from scientific papers.

3.2. Comparative Analysis

1. Plot the degree distribution (PMF and CCDF) for the Citation network on log-log scales.
2. Use the `powerlaw` library to fit a power-law and estimate its γ.
3. Compare the estimated γ from the real network with the one from your BA model.

Answer the following questions in your lab report:

1. Does the degree distribution of the Citation network follow a power-law? Over what range (`xmin`)?
2. How does the estimated γ for the Citation network compare to the theoretical value from the BA model? What might account for any differences?
3. The BA model incorporates only *growth* and *preferential attachment*. What are some real-world social processes that these mechanisms might represent? What important social processes are *missing* from this simple model? (e.g., homophily, aging of nodes, external events).